# Abhishek Sudarshan

## System Development Engineer 2 @ Amazon

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### SUMMARY

System Development Engineer 2 at Amazon specializing in distributed event-driven systems on AWS. Focus on JVM/Kotlin serverless performance, multi-day Step Functions orchestration, bounded concurrency, single-retry ownership per failure domain, zero-trust security migrations, and safe progressive deployment.

### EXPERIENCE

**System Development Engineer 2** — Amazon 2021 – Present | India

- **Zero-Trust Storage & Access:** Designed and executed phased migration to identity-bound OIDC access with server-side tenant isolation, safe retries, and bounded memory caps (zero pre-production errors, eliminated silent data corruption).
- **Serverless Fan-Out & Cold Starts:** Profiled and tuned JVM Lambda cold starts from 8–11s to ~1.5s (P99 latency −70%); implemented claim-check pattern on S3/SQS for large payloads and redesigned retry budgets to absorb bursts without quota increases.
- **Self-Healing Workflows:** Instituted single-retry ownership per failure class with stall alarms in Step Functions; built idempotent bulk-recovery CLI (redrive, step restart, dry-run) cutting recovery from 6–8 hours to 10–15 minutes (MTTR −70%, ~15% compute saved on Spot).
- **Scalable Data Access:** Engineered backward-compatible `search_after` cursor pagination across OpenSearch with sort tie-breakers and conditional DynamoDB writes (~65% faster lookups, zero integrity failures).
- **Progressive Delivery & Safety:** Established team delivery standards (bake time, composite alarms, automated rollback, CodeDeploy gradual shifting, synthetic canaries), increasing full-CD from 48% to 84% and cutting lead time from 53 to 16 days.
- **Bedrock RAG Assistant:** Architected engineering assistant on Amazon Bedrock Knowledge Bases with intent-based routing; improved accuracy from ~60% to ~95% across 25+ adopted teams.

**Software Engineer (TCS Digital)** — Tata Consultancy Services June 2019 – 2021 | India

- **Data Migration Engine:** Built high-concurrency batch migration modules orchestrating multi-tenant branch transfers via REST APIs with automated reconciliation and zero data loss.
- **Banking Modernization:** Delivered Spring Boot microservice & Liferay DXP Proof-of-Concepts for a tier-1 European banking client, modernizing core monolith flows.

### SELECTED PRODUCTION SYSTEMS

**Zero-Trust Storage Migration** `Zero Errors • OIDC`

Replaced pre-signed URLs with identity-bound OIDC auth, KMS encryption context, and memory caps.

**Serverless Fan-Out at Scale** `Cold Start 1.5s • P99 −70%`

Claim-check on S3/SQS for large messages, provisioned concurrency, and token-bucket retry budgets.

**Bulk Recovery CLI** `Recovery 10-15m • MTTR −70%`

Idempotent state redrive CLI, stall detection alarms, and single-retry ownership on Step Functions.

**Cursor-Paginated History API** `~65% Faster • TOCTOU-Safe`

Backward-compatible pagination contract with sort tie-breakers on OpenSearch & DynamoDB.

### SKILLS & COMPETENCIES

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|---|---|
| **Distributed Systems:** | Event-driven fan-out, claim-check, jittered exponential backoff, Step Functions, token bucket, multi-tenant OIDC isolation |
| **AWS Cloud Platform:** | Lambda, ECS Fargate, EC2 Spot, SQS, SNS, Step Functions, S3, API Gateway, DynamoDB, OpenSearch, KMS, IAM, VPC |
| **Languages & Core:** | Kotlin (primary), Java, TypeScript, Python, Smithy models, Concurrency Control, Coroutines, MockK, JUnit 5 |
| **AI & LLM / DevOps:** | Amazon Bedrock Knowledge Bases (RAG), Converse API, AWS CDK, CodeDeploy Gradual Shifting, Composite Alarms, Docker |

### EDUCATION

**B.S. in Computer Science & Engineering** — Gujarat Technological University 2015 – 2019 | India